

ACRONYMS, ABBREVIATIONS AND SYMBOLS

A	Area of the catchment (km ²)
A'	Area of the reservoir at full supply level (ha or m ²)
D	Maximum depth of spillway at the crest
DFID	UK Department for International Development
FSL	Full supply level
L	Length of the dam at full supply level
H	Crest height of dam at full supply level (m)
H'	Maximum height of dam at full supply level (m)
H''	Freeboard height (m)
NPC	National Population Commission
PMF	Probable maximum flood
Q	Dam capacity (m ³)
T	Throwback (m)
USDA	United States Department of Agriculture
Y	Catchment yield (m ³)

INTRODUCTION

The design of an earth fill (modified) dam was undertaken with location at the Mgbowo Community. The dam project is expected to;

- Supply water to the five villages that make up the Mgbowo community both for communal purposes
- Serve also as a mini hydropower generator for productive use by the community which has recently been built an ICT centre and also for the agro processing unit in the Community Development Centre.
- Ensure that there is enough soil moisture for round the year farming via irrigation schemes that would be tied to the construction of the dam.

Earth embankments have been used since the earliest times to impound and divert water. They are simple compacted structures that rely on their mass to resist sliding and overturning and are the most common type of dam found worldwide. Modern haulage methods and developments in soil mechanics since the end of the nineteenth century have greatly increased the safety and life of these structures.

BACKGROUND

DESK STUDY

Location

Mgbowo is located in Agwu LGA in Enugu State. It is situated at Latitude 06°05'N and Longitude 07°31' E. it is south of Enugu and about 72km from Amaraka, along the Enugu-Umahia Road (A3).



Plate 1: Satellite imagery showing the actual location of Mgbowo with Grid Lines (Source: Google Earth)

There are five villages that make up Mgbowo Community, they include; Ezioha, Amata, Imeana, Inyi and Eziobodo,. The people are predominantly farmers and rely on the yearly farm harvest for sustenance and very little do farming on a large scale. Major crops include maize, yam tubers, cassava etc.



Plate 2: Satellite Imagery modified to show the five villages that make up the Mgbowo community

Population

The Population of the area as at the 1991 census was estimated to be 13,293 while that of the Awgu LGA as at the 2006 Census was estimated to be 390,681 persons. It is densely populated with 966 people per square mile.

Geology of the Area

The area is located amidst the Agwu shales of coincian age, bituminous coals with oil deposits, it is classed under the Southern -Nigerian Sedimentary Basin, under the Agwu formation with thickness of about 10m in some locations.

Topography

The mgbowo region is placed in the region of enugu of Nigria with an average elevation of 325ft (99.05m) above sea level. Hilly in some locations the whole area is predominantly plain with major areas lying around 50m from sea level.

Climatic Information

The Mgbowo Climate is majorly tropical rain forest which makes rainfall one of the most important climatological elements in the area. Although humid, a very dry period in the area can result in crop failures, food deficits and in the extreme, starvation. The mean annual precipitation is calculated to be 1950mm/year, with temperature ranging from about 20°C in december to 34.7°C in march.

DEMOGRAPHIC STUDIES

DATA OBTAINED

The following population data were obtained from the National Population Commission for censuses conducted during the 1991 and 2006 censuses respectively;

Table 1: Population Statistics Obtained for Agwu LGA and Mgbowo (Source: NPC)

Year	AGWU LGA	MGBOWO (Calculated)
1991	232,245	13,293
2006	390,681	-

POPULATION GROWTH RATE

Definition of Population growth rate: The average annual percent change in the population, resulting from a surplus (or deficit) of births over deaths and the balance of migrants entering and leaving a country.

Using the obtained population counts obtained for the period, we were able to calculate the growth rate within the period using the formulae as follows;

$$P = A(1 + i)^t$$

Where;

P = Population as at 2006 (NPC, 2006 Census)

A = Population as at 1999 (NPC, 199 Census)

i = Growth Rate

t = Number of years between 1999 and 2006

$$390681 = 232245(1 + i)^{15}$$

$$\frac{390681}{232245} = (1 + i)^{15}$$

$$1.68 = (1 + i)^{15}$$

$$\sqrt[15]{1.68} = (1 + i)$$

$$1.0352 = (1 + i)$$

$$i = 1 - 1.0352$$

$$i = 0.0352$$

In percentage,

$$i = 3.52\%$$

Utilizing the calculated growth rate data obtained, we calculated the year population increase till 2012 using the formulae

$$P = A(1 + i)$$

Where,

P = Estimated population of current year

A = Population at previous year

i = Growth Rate

t was taken as 1 because of the the one year incremental change in time.

Table 2: Calculated Population Estimates

Year	Growth Rate (%)	Population (Estimated from 2006)	
		AGWU LGA	MGBOWO
2006	-	390,681	22362
2007	3.52	404,433	23149
2008	3.52	418669	23964
2009	3.52	433,406	24808
2010	3.52	448662	25681
2011	3.52	464455	26585
2012	3.52	480804	27521

POPULATION PROJECTIONS

Agwu LGA Community

The dam is projected to be built for a 50 year life for utilization by the agwu local government area, so therefore our projections for population would cover the same.

Taking a geometric increase in population growth (3.25%, 2006), and using the formulae;

$$P = A(1 + i)^t$$

Where,

P = Estimated population at final year

A = Population at base year

i = Growth Rate

t = number of years after base year

Population after 50 years (i.e A.D 2063)

$$P = A(1 + i)^t$$

$$P_{50} = 480804(1 + 0.0352)^{50}$$

$$P_{50} = 480804(1.0352)^{50}$$

$$P_{50} = 2,711,323 \text{ People.}$$

HYDROLOGICAL STUDIES

WATER DEMAND

Water demand is defined as the volume of water requested by users to satisfy their needs. In a simplified way it is often considered equal to water consumption, although conceptually the two terms do not have the same meaning. This is because in some cases, especially in rural parts of southern Africa, the theoretical water demand considerably exceeds the actual consumptive water use.

Irrigation water demand

The irrigation water demand and use can be estimated approximately at a catchment and sub catchment level using the following equations:

$$\text{Irrigation water demand } IR = (ET_o - P_e)A$$

Where:

ET_o is the reference crop evapotranspiration for grass;

P_e is the effective rainfall;

A is the area under irrigation.

$$\text{Irrigation water use} = \frac{IR}{E_i}$$

Where:

IR is the irrigation water demand;

E_i is the overall irrigation efficiency.

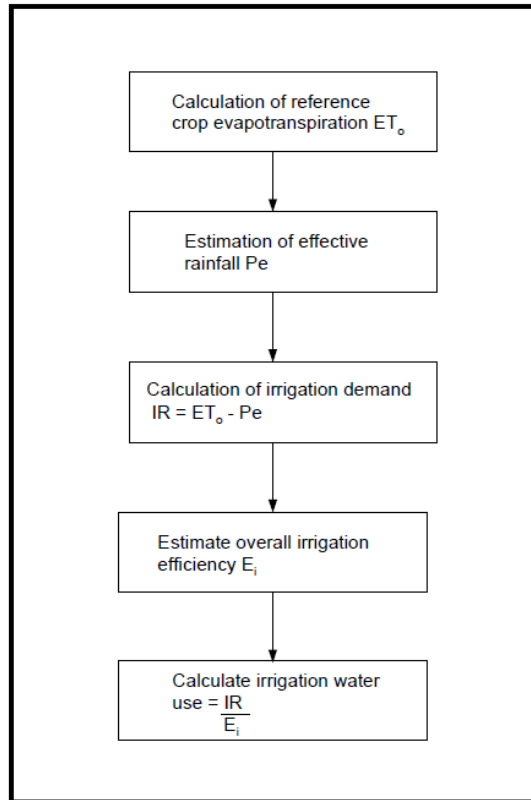


Fig 1: Approximate Method for Estimating Irrigation Demand and Use.

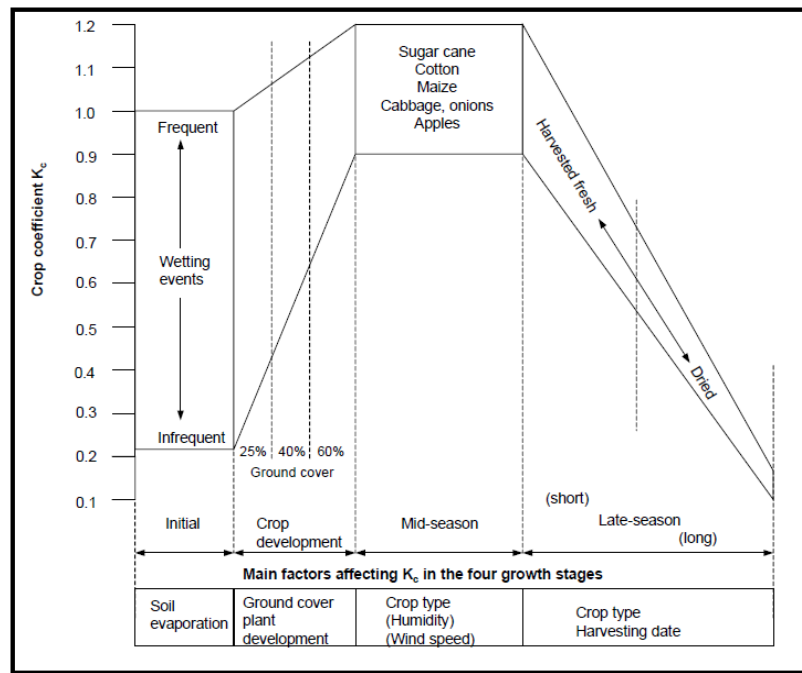
Estimation of the crop coefficient

Crop evapotranspiration is the amount of water utilised by a crop under standard conditions i.e. a disease free, well fertilised crop grown in a large field under optimum soil conditions (References 4.9 and 4.10). The crop evapotranspiration ET_{crop} in mm/day is calculated by multiplying the reference crop evapotranspiration ET_o in mm/day by a dimensionless crop coefficient K_c :

$$ET_{crop} = K_c ET_o$$

The crop coefficient is dependent upon the following factors:

- Crop type;
- Climate;
- Soil evaporation;
- Crop growth stage.



Source: Reference 4.9

Fig. 2: Design Curves for Determining Crop Coefficient (Doorenbos and Pruitt (1977))

We took a Conservative value of 1.2 for our crop coefficient

Estimation of effective rainfall

Crop water requirements may be fully or partly met by precipitation. However, it should be noted that not all rainfall is effective as some may be lost to surface runoff, deep percolation or evaporation. The effective rainfall may be defined as the proportion of the rainfall that is useful for crop production. There are several factors that affect effective rainfall including:

- Rainfall characteristics;
- Soil type;
- Groundwater characteristics;
- Management practices.

There are several empirically based methods available to calculate effective rainfall including:

- Renfro equation;
- US Bureau of Reclamation method;
- US Department of Agriculture Soil Conservation Service method.

The US Department of Agriculture Soil Conservation Service method is often used. The table below allows the mean monthly effective rainfall to be estimate provided the crop evapotranspiration and mean monthly rainfall are known.

Table 3: Mean monthly effective rainfall related to Mean Monthly ET_{crop} and Mean Monthly Rainfall

Mean monthly rainfall (mm)		12.5	25	37.5	50	62.5	75	87.5	100	112.5	125	137.5	150
		Mean monthly effective rainfall (mm)											
Mean monthly ET_{crop} (mm)	25	8	16	24									
	50	8	17	25	32	39	46						
	75	9	18	27	34	41	48	56	62	69			
	100	9	19	28	35	43	52	59	66	73	80	87	94
	125	10	20	30	37	46	54	62	70	76	85	92	98
	150	10	21	31	39	49	57	66	74	81	89	97	104
	175	11	23	32	42	52	61	69	78	86	95	103	111
	200	11	24	33	44	54	64	73	82	91	100	109	117
	225	12	25	35	47	57	68	78	87	96	106	115	124
	250	13	25	38	50	61	72	84	92	102	112	121	132

Using the rainfall data obtained and the conservative assumptions above, we obtained the following table;

Table 4: Calculated Irrigation Water Demand and Use

Month	Reference crop evapotranspiration (mm)	Average Rainfall	Effective rainfall (mm)	Irrigation Requirement	Irrigation water demand (million m ³)	Irrigation water use (million m ³)
January	124	0.0	0	124	1.24	1.55
February	109	2.0	0	109	1.09	1.36
March	127	43.5	37	90	0.9	1.13
April	117	96.8	66	51	0.51	0.64
May	115	168.4	94	21	0.21	0.26
June	102	265.2	98	4	0.04	0.05
July	112	232.4	124	0	0	0.00
August	143	340.4	132	11	0.11	0.14
September	177	277.2	117	60	0.6	0.75
October	202	170.2	132	70	0.7	0.88
November	159	40.9	10	149	1.49	1.86
December	133	7.3	9	124	1.24	1.55
				Total	8.13	10.16

LIVESTOCK DEMAND

Other uses such as livestock water can be calculated by estimating water use for this. FAO can provide advice as well as locally based government and other organizations. As a guideline the following (assuming the animals are on dry pastures and good quality water is available) can be used:

- Cattle 40-80 litres/day for each animal (milking cows may need 100 litres/day).
- Young stock 25-50 litres/day.
- Pigs 25 litres/day.
- Poultry 30 litres/day per 100 adult birds.
- Sheep 2-6 litres/day.
- Goats 3-8 litres/day.

10 percent was added to the calculated total amount for water use by wild and feral animals.

Assuming the following;

0.05	Cattle per Household
0.25	Young Stock per Household
0.15	Pigs per household
10	Poultry Per Household
0.35	Goats per Household
0.15	Sheep per Household

And taking a average number of people per household to be 8 per household.

Therefore;

Number of House holds equals;

$$\begin{aligned} & \frac{\text{Total Population}}{\text{Number per Household}} \\ &= \frac{2711232}{8} \\ &= 338904 \text{ Households} \end{aligned}$$

Therefore calculating for the livestock water demand we have,

Table 5: Calculated Livestock Water Demand

LIVESTOCK		TOTAL NUMBER	LITRES/DAY	TOTAL LITRES/DAY
Cattle	0.05	16945.2	80	1355600
Young Stock	0.25	84726	50	4236300
Pigs	0.15	50836	25	1270900
Poultry	1	338904	0.3	101671
Goats	0.35	118616	6	711696
Sheep	0.15	50836	8	406688
TOTAL				8082855

DOMESTIC WATER DEMAND

This includes water for drinking, washing, cooking and other general household activities.

This demand is estimated to be 120 l/cap/day.

COMMERCIAL AND INDUSTRIAL DEMAND

This takes into consideration various water demand in our Commercial/Industrial centres. It is estimated at 200 l/p/d.

Industrial demand – 50 l/p/d

Health Centre – 50 l/p/d

Schools – 20 l/p/d

Commercial Centre – 50 l/p/d

Churches and Mosques – 30 l/p/d

FIRE DEMAND

This is a reserved quantity of water kept should in case there exist a fire outbreak. It is estimated at 20 l/p/d.

PUBLIC USE

This is the quantity of water used for diverse public works such as cleaning of sewers, watering of lawns (gardens), washing of public parks, etc. this demand is estimated as 70 l/p/d.

TOTAL WATER DEMAND

Using all the water demand requirements assumed and obtained we then proceeded to calculate the total water demand.

Domestic Demand	– 120 l/p/d
Commercial and Industrial demand	– 200 l/p/d
Fire Demand	– 20 l/p/d
Public Use	– 70 l/p/d
Total Human Demand demand (a)	= 410 l/p/d

Population as at year 2012 = 480,804 persons

Projected Population for year 2063 = 2,711,237 persons (Based on 50 Year projection)

$$\begin{aligned}\text{Required Quantity of water (a)} &= 410 \times 365 \times 2,711,237 \\ &= 4.05 \times 10^{11} \text{ L} \\ &= 4.05 \times 10^8 \text{ m}^3 \\ &= \mathbf{405 \text{ million m}^3}\end{aligned}$$

$$\text{Livestock Demand} \quad \mathbf{8.08 \times 10^{-3} \text{ million m}^3}$$

$$\text{Irrigation Demand} \quad \mathbf{10.16 \text{ million m}^3}$$

$$\text{Total Water Demand} \quad \mathbf{= 415.17 \text{ million m}^3}$$

CATCHMENT YIELD

The catchment yield, 'Y', is based on the expected annual runoff from a catchment and is an important factor in assessing the feasibility of a dam and in determining the required height of the embankment. This is important to allow for sizing the dam to suit expected inflow and estimate the area that can be irrigated. It is estimated as follows:

The average percentage of runoff is not known, so a figure of 10 percent of the mean annual rainfall for the catchment area is utilized for calculating the runoff.

The annual runoff for the catchment, in mm, based on the percentage determined is 'Rr'.

Since Mean annual Rainfall is 137.02mm,

Therefore

$$\begin{aligned}\text{Runoff } R_r &= 10\% \text{ of Mean annual rainfall} \\ &= 137.02 \times 0.1 \\ &= 13.702 \text{ mm}\end{aligned}$$

The catchment area 'A' in km², upstream of the proposed embankment is estimated to be equal to

The annual runoff for the catchment (the catchment yield in an average year),

Y, in m³, is given by:

$$\begin{aligned}Y &= R_r \times A \times 1000 \\ Y &= 13.702 \times\end{aligned}$$

CALCULATING THE PMF USING THE RATIONAL METHOD

The procedure to follow in calculating the probable maximum flood using the Rational Method is:

1. Locate the dam or new site on the appropriate topographic map (1:50 000 scale is normally suitable for all but the smallest catchments) and draw on the catchment boundary upstream of the embankment centre-line. Using a planimeter measure the catchment area, 'A', in km².
2. Using a linear measuring wheel or similar device, measure the 'actual length, l', of the main river/stream, upstream of the site and to the main river source, in km.

3. Estimate the elevation difference, 'h', in m, between the dam site and the 11 Usually 1 in 20, 25 or 50 years for small dams but it can be as high as 1 in 1 000 years for larger dams.

In the case of Kariba dam, the spillway is designed for a 1 in 10 000 PMF.

5 Site selection and preliminary investigations **37** main stream at its source. A contour map is essential for this and some extrapolation may be required where contour intervals are large.

4. Determine the time of concentration, 'Tc', in hours, using the formula:

$$T_c = (0.87 L^3/h)^{0.385}$$

5. From rainfall records or a rainfall distribution map, estimate the mean annual rainfall for the catchment. Using a graph similar to **Figure 6a**, estimate the one day storm rainfall, 'P', for the selected return period¹². Use 1:20 to 1:25 year return periods for smaller catchments and 1:50 year return periods for larger dams, larger catchments or dams where safety issues are more important (i.e. near populated areas).

6. Derive the storm depth ratio, 'R', from the graph in **Figure 6b** and using the Tc determined above.

7. Calculate the extreme height channel slope as a percentage [100 h/(1 000 litres)] and estimate a runoff coefficient, "Cr", for the assumed return period using the graph in **Figure 6c**. If runoff is known to be excessive, such as on bare, eroded slopes, the runoff coefficient can be increased by up to 20 percent.

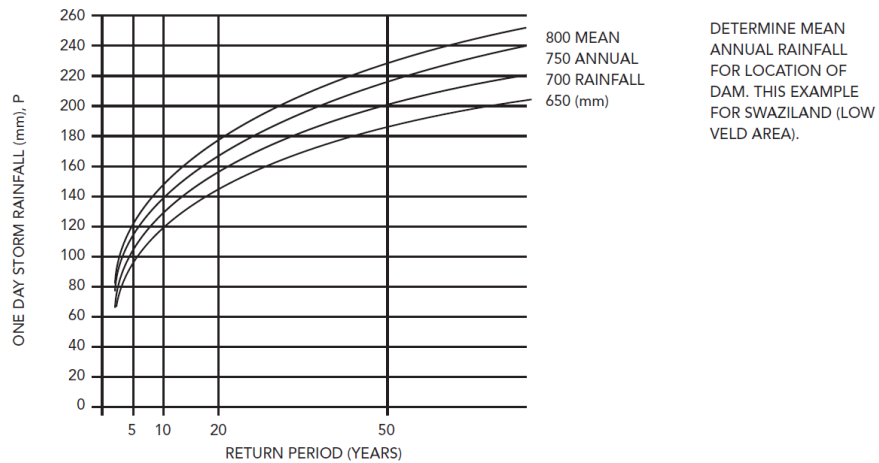
8. Determine the probable maximum flood (PMF), 'Qp', in m³/s, using:

$$Q_p = 0.278 A P R C_r / T_c$$

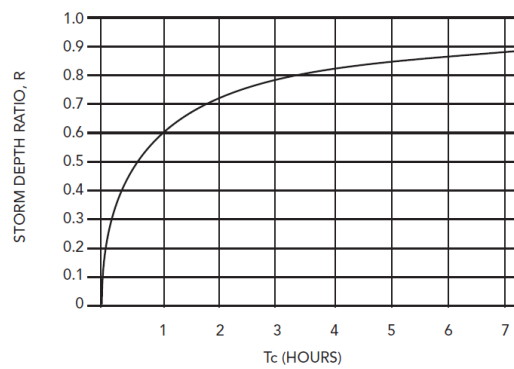
Where no other data are available and figures such as 6a to 6b cannot be drafted, a very approximate peak flood estimate can be made by taking the highest daily rainfall figure for the catchment and assuming that all dams in the same catchment are 100 percent full, the ground is saturated and that 100 percent runoff will occur.

For example, if a rainfall of 223 mm fell on a catchment area of 19 km², the estimated peak flood would be in the region of 49 m³/s over a 24 h period. Always stay on the conservative side when using approximations or estimates for peak floods;

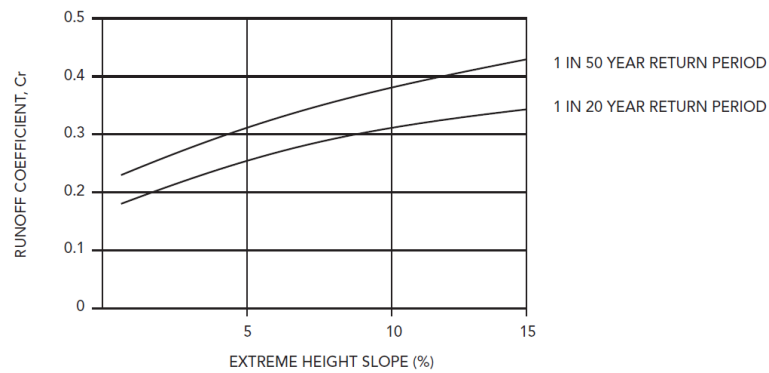
2-4 m³/s per km² of catchment area per 24 h period is a guide but this figure should always be adapted bearing in mind local topographic and climatic conditions.



ONE DAY STORM RAINFALL



STORM DEPTH RATIO



ESTIMATES OF STORAGE REQUIRED

DAM DESIGN

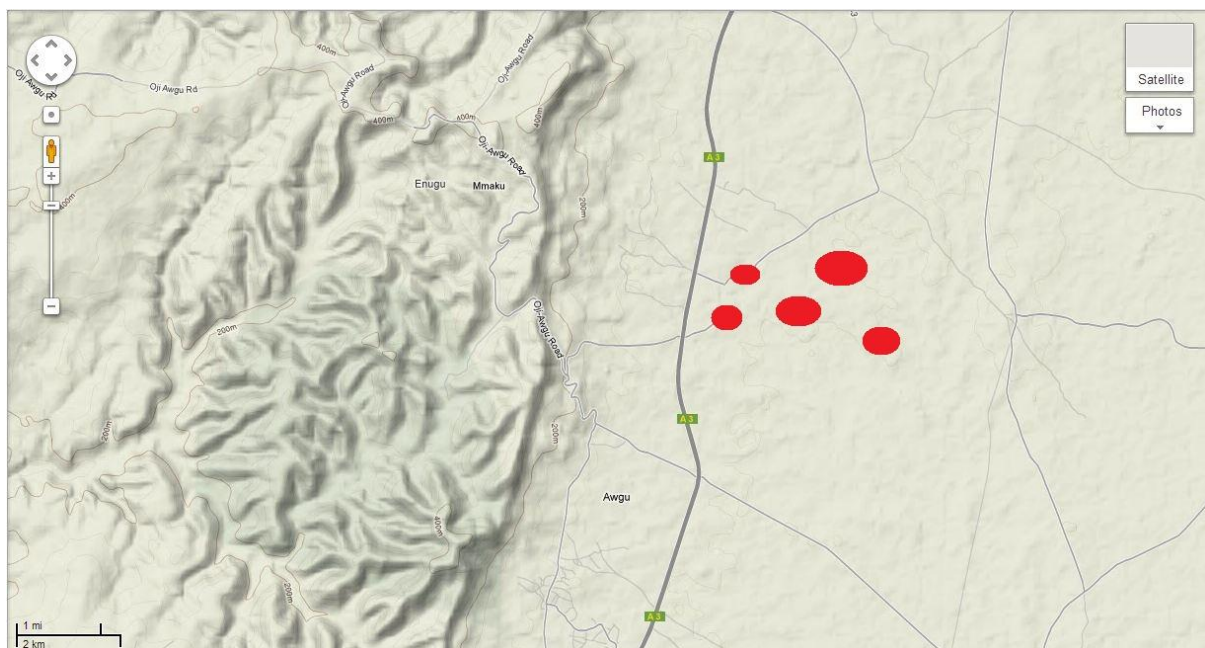
HAZARD POTENTIAL

Because of the location of the community, an assessment of the hazard potential was done for the appropriate location of the dam to ensure that the location of the dam does not in any way cause loss of life and untold economic loss in the case of failure. Table 1 provides guidelines that were utilized to characterize the possible locations and their hazard potential.

Table 1: Hazard potential

Loss of life	Economic loss	Hazard potential
Almost impossible	Negligible	None
Extremely unlikely	Minimal	Very low
Improbable	Marginal	Low
Possible	Appreciable	Moderate
Probable	Excessive	High

Locations that would place the community itself on the downstream were categorized as high risk zones and were not considered for the siting of a dam, it was decided that the dam be sited in such a way that the communities would lie on the upstream of the dam with the spill way providing appropriate levelling of the water body ensuring that it doesn't flood the community.



SELECTION CRITERIA

The following criteria were used for the selection of the type of dam to be constructed and used;

1. Topography: the topography of the selected area
2. The main advantages involved in the construction of small earth dams are:
3. __ Local natural materials are used.
4. __ Design procedures are straightforward.
5. __ Comparatively small plant and equipment are required.
6. __ Foundation requirements are less stringent than for other types of dam. The
7. broad base of an earth dam spreads the load on the foundation.
8. __ Earthfill dams resist settlement and movement better than more rigid
9. structures and can be more suitable for areas where earth movements are
10. common.

However, disadvantages also exist and these are:

__ An earth embankment is easily damaged or destroyed by water flowing on, over or against it. Thus, a spillway and adequate upstream protection are essential for any earth dam.

__ Designing and constructing adequate spillways is usually the most technically difficult part of any dam building work. Any site with a poor quality spillway should not be used.

__ If not adequately compacted during construction, the dam will offer weak structural integrity, offering possible pathways for preferential seepage.

__ Earth dams require continual maintenance to prevent erosion, tree growth, subsidence, animal and insect damage and seepage.

They are,

however, generally inferior to the modern method of zoned construction in which an embankment is built in three sections:

__ upstream and relatively impermeable section;

__ central core or hearting of highly impermeable material (which, with any below ground cutoff, will effectively seal the dam against seepage); and

__ downstream section of poorer, coarser material that allows freer drainage of the structure and which, by its weight, anchors the complete embankment to its foundation and prevents slip and other movement.

FREEBOARD

Freeboard for small dams should never be less than 0.5 m with xx m to xx m preferred.

Where wave action is likely, additional freeboard may be required. This can be estimated using the following formula:

$$\text{Freeboard height, } H'' \text{ (in m)} = 0.014 (F)^{0.5}$$

Where H'' is the freeboard height and F is the fetch which is the longest distance, in km, across the storage area (usually measured in a straight line from the centre line of the proposed embankment to the tail water area of the proposed reservoir).

The overall freeboard height can then be calculated taking into account the wet freeboard, H'' , (as estimated with the formula above) required to counteract wave action and the dry freeboard (estimated by the engineer) for safety and other factors. The total freeboard is effectively the design depth for the spillway (at its entrance).

CORE DEPTH AND THICKNESS

CREST WIDTH

The crest width of an embankment is selected taking into account the size of the dam, the catchment characteristics and topography and whether road or other access will be required across the embankment. In all cases, the embankment crest width should be designed to allow the safe passage of plant and equipment to be used in the dam construction and should be no less than 2 m wide.

A standard crest width using the formula below was used:

$$Cw \text{ (in m)} = 0.4H + 1$$

Where,

Cw is the crest width and

H is the maximum height of the dam in metres.

To reduce erosion, the crest was given a 2.5 percent cross fall to drain rainwater to the reservoir via the upstream slope of the embankment.

PRELIMINARY VOLUME OF EARTHWORKS

The volume of earthwork can be estimated as follows:

$$V = 0.216 HL (2C + HS)$$

Where:

V is the volume of earthworks in m^3 .

H is the crest height (FSL+ freeboard) of the dam in m.

L is the length of the dam, at crest height H , in m (including spillway).

C is the crest width in m.

S is the combined slope value.

For example, if the slopes of the embankment are 1: 2 and 1:1.75, $S = 3.75$.

STORAGE CAPACITY

At this stage, the storage capacity is worked out as follows:

$$Q = 6LTH'$$

Where:

Q is the storage capacity in m³ and should not exceed Y above.

L is the length of the dam wall at full supply level (FSL) in m.

T is the throwback, in m and approximately in a straight line from the wall.

H' is the maximum height of the dam, in m, at FSL.

6 is a factor (conservative generally) that can be adjusted (to 5 or 4) with experience and local knowledge.

Q=

RUN OFF COEFFICIENT

Once the PMF has been estimated, the spillway width can be calculated using the formula:

$$Q_p = 1.7 b D^{1.5}$$

Where b and D are in m and Q_p is in m³/s

'1.7' is a factor derived for concrete ogee type crests and can vary up to 2.25 according to site conditions and factors of safety. 1.7 is generally used for spillways for small dams on small catchments.

'b' is the minimum width ('breadth') of the spillway and is calculated by introducing the values for Q_p (estimated using the options above) and D^{1.5}. It is assumed that b is large when compared to D and that the spillway channel will thus be rectangular.

'D' is the depth of the spillway at the crest and will comprise all or part of the design freeboard. D is normally in the range 0.75 m to 1.5 m for small dams and comprises the total freeboard. However, where wave action or backing up of floods may affect the dam, an additional 'dry' freeboard of up to 0.75 m. should be added to the figure above for safety reasons.

Once all the other values are known, 'b' can then be calculated and the best option for varying depths, 'D', can be chosen.

The width 'b' is the minimum width for the spillway to accommodate the design flood. It assumes that there is no constriction downstream of the spillway. The width and depth may

have to be adjusted to suit the local topography and spillway bed material later in the design process.